

COIT20246 Cyber Security and Networking

Project Specification 1

**Small Business Network Security, OpenWrt Firewall Configuration and Cyber
Security Risk Assessment**

Student 1: Pavan Reddy Chinthalapally

Student ID: 12327397

Student 2: Patel Ronak Ghanshyambhai

Student ID: 12328755

**Business Scenario: Small IT and Business Consultancy, Brisbane,
Queensland, Australia**


```
R23 [Running] - Oracle VirtualBox
File Devices Help

root@OpenWrt:~# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master br-mng
    state UP qlen 1000
    link/ether 08:00:27:8e:c5:ff brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP qlen
    1000
    link/ether 08:00:27:a8:f2:21 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fea8:f221/64 scope link
        valid_lft forever preferred_lft forever
4: br-mng: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP qlen
    1000
    link/ether 08:00:27:8e:c5:ff brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.2/24 brd 192.168.56.255 scope global br-mng
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe8e:c5ff/64 scope link
        valid_lft forever preferred_lft forever
root@OpenWrt:~#
```

Figure 2: OpenWrt Network Interface Configuration and IP Address Allocation

```
ip addr add|del IFADDR dev IFACE ; show|flush [dev IFACE] [to PREFIX]
ip route list|flush|add|del|change|append|replace|test ROUTE
ip link set IFACE [up|down] [arp on|off] [multicast on|off]
    [promisc on|off] [mtu NUM] [name NAME] [qlen NUM] [address MAC]
    [master IFACE ; nomaster] [netns PID]
ip neigh show|flush [to PREFIX] [dev DEV] [nud STATE]
ip rule [list] ; add|del SELECTOR ACTION
root@OpenWrt:~# ubus call system board
{
    "kernel": "5.10.161",
    "hostname": "OpenWrt",
    "system": "11th Gen Intel(R) Core(TM) i5-11400F @ 2.60GHz",
    "model": "innotek GmbH VirtualBox",
    "board_name": "innotek-gmbh-virtualbox",
    "rootfs_type": "ext4",
    "release": {
        "distribution": "OpenWrt",
        "version": "22.03.3",
        "revision": "r20028-43d71ad93e",
        "target": "x86_64",
        "description": "OpenWrt 22.03.3 r20028-43d71ad93e"
    }
}
root@OpenWrt:~#
```

Figure 3: OpenWrt System and Virtual Machine Information

```
valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:fea8:f221/64 scope link
valid_lft forever preferred_lft forever
4: br-mng: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP qlen 1000
    link/ether 08:00:27:8e:c5:ff brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.2/24 brd 192.168.56.255 scope global br-mng
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fea8:c5ff/64 scope link
        valid_lft forever preferred_lft forever
root@OpenWrt:~# ip -br addr
BusyBox v1.35.0 (2023-01-03 00:24:21 UTC) multi-call binary.

Usage: ip [OPTIONS] address|route|link|neigh|rule [ARGS]

OPTIONS := -f[amily] inet|inet6|link | -o[neline]

ip addr add|del IFADDR dev IFACE | show|flush [dev IFACE] [to PREFIX]
ip route list|flush|add|del|change|append|replace|test ROUTE
ip link set IFACE [up|down] [arp on|off] [multicast on|off]
    [promisc on|off] [mtu NUM] [name NAME] [qlen NUM] [address MAC]
    [master IFACE | nomaster] [netns PID]
ip neigh show|flush [to PREFIX] [dev DEV] [nud STATE]
ip rule [list] | add|del SELECTOR ACTION
root@OpenWrt:~#
```

Figure 4: OpenWrt Network Configuration and Interface Mapping

```
network.loopback.ipaddr='127.0.0.1'
network.loopback.netmask='255.0.0.0'
network.globals=globals
network.globals.ula_prefix='fd88:3563:c944::/48'
network.@device[0]=device
network.@device[0].name='br-lan'
network.@device[0].type='bridge'
network.@device[0].ports='eth0'
network.wan=interface
network.wan.device='eth1'
network.wan.proto='dhcp'
network.wan.ifname='eth1'
network.mng=interface
network.mng.proto='static'
network.mng.netmask='255.255.255.0'
network.mng.ipaddr='192.168.56.2'
network.mng.device='br-mng'
network.lan=interface
network.lan.proto='dhcp'
network.lan.device='eth2'
network.@device[1]=device
network.@device[1].name='br-mng'
network.@device[1].type='bridge'
network.@device[1].ports='eth0'
root@OpenWrt:~#
```

Figure 5: OpenWrt Network Interface and DHCP Configuration

```

network.mng.device='br-mng'
network.lan=interface
network.lan.proto='dhcp'
network.lan.device='eth2'
network.@device[1]=device
network.@device[1].name='br-mng'
network.@device[1].type='bridge'
network.@device[1].ports='eth0'
root@OpenWrt:~# ip route
default via 10.0.3.2 dev eth1 src 10.0.3.15
10.0.3.0/24 dev eth1 scope link src 10.0.3.15
192.168.56.0/24 dev br-mng scope link src 192.168.56.2
root@OpenWrt:~# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master br-mng
    state UP qlen 1000
    link/ether 08:00:27:8e:c5:ff brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP qlen
    1000
    link/ether 08:00:27:a8:f2:21 brd ff:ff:ff:ff:ff:ff
4: br-mng: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP qlen
    1000
    link/ether 08:00:27:8e:c5:ff brd ff:ff:ff:ff:ff:ff
root@OpenWrt:~# _

```

Figure 6: OpenWrt Routing Table and Network Interface Status

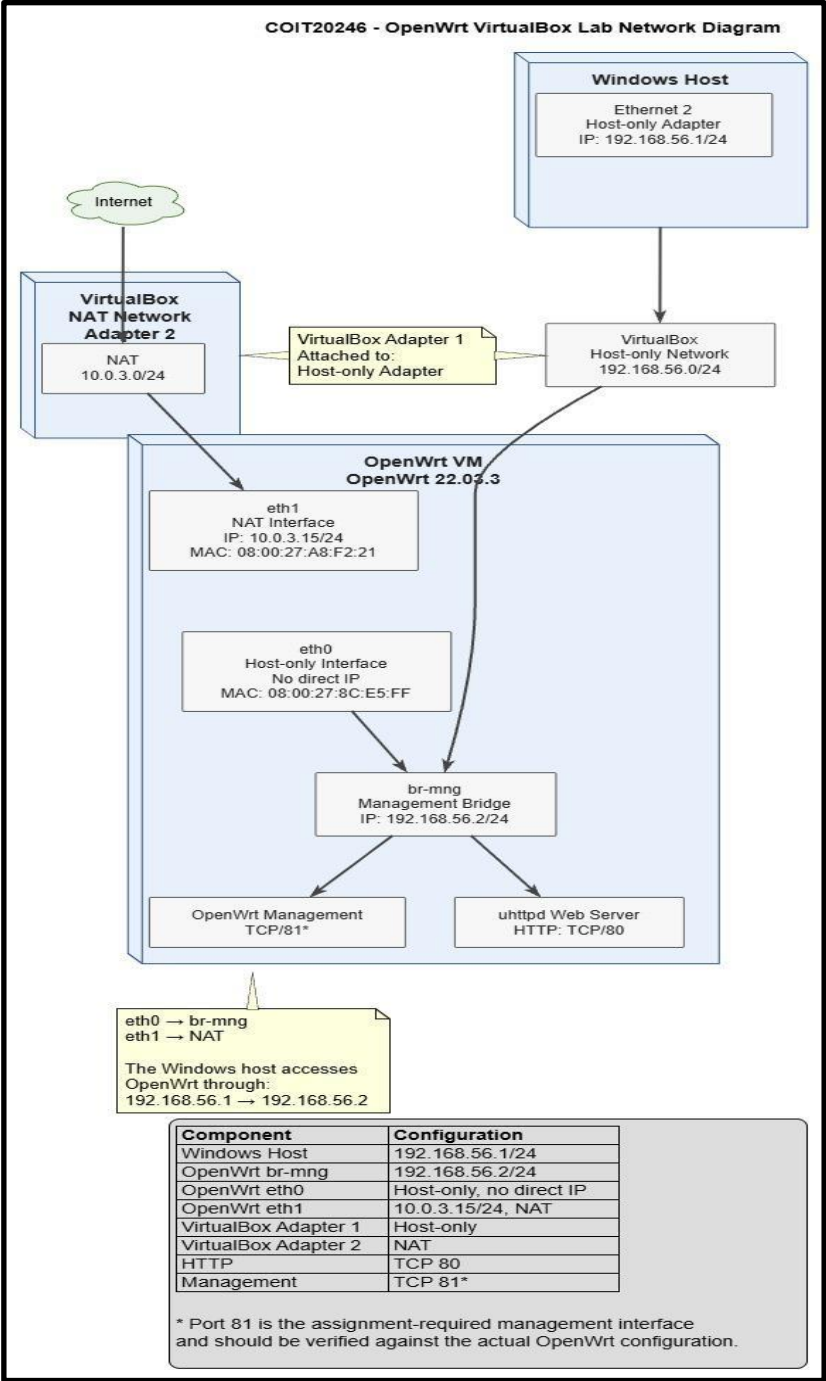


Figure 7: Proposed OpenWrt VirtualBox Laboratory Network Topology and IP Address Allocation

```
C:\Users\admin>ping 192.168.56.2

Pinging 192.168.56.2 with 32 bytes of data:
Reply from 192.168.56.2: bytes=32 time<1ms TTL=64
Reply from 192.168.56.2: bytes=32 time<1ms TTL=64
Reply from 192.168.56.2: bytes=32 time<1ms TTL=64
Reply from 192.168.56.2: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.56.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\admin>
```

Figure 8: Successful ICMP Connectivity Test from Windows Host to OpenWrt

```
root@OpenWrt:~#
root@OpenWrt:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: seq=0 ttl=255 time=27.300 ms
64 bytes from 8.8.8.8: seq=1 ttl=255 time=27.892 ms
64 bytes from 8.8.8.8: seq=2 ttl=255 time=26.690 ms
64 bytes from 8.8.8.8: seq=3 ttl=255 time=28.318 ms
64 bytes from 8.8.8.8: seq=4 ttl=255 time=27.975 ms
64 bytes from 8.8.8.8: seq=5 ttl=255 time=27.402 ms
^C
--- 8.8.8.8 ping statistics ---
7 packets transmitted, 6 packets received, 14% packet loss
round-trip min/avg/max = 26.690/27.596/28.318 ms
root@OpenWrt:~# ping google.com
PING google.com (192.178.158.100): 56 data bytes
64 bytes from 192.178.158.100: seq=0 ttl=255 time=28.663 ms
64 bytes from 192.178.158.100: seq=1 ttl=255 time=28.525 ms
64 bytes from 192.178.158.100: seq=2 ttl=255 time=28.480 ms
64 bytes from 192.178.158.100: seq=3 ttl=255 time=30.433 ms
64 bytes from 192.178.158.100: seq=4 ttl=255 time=29.078 ms
^C
--- google.com ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 28.480/29.035/30.433 ms
root@OpenWrt:~#
```

Figure 9: Successful Internet Connectivity Test from OpenWrt to External Networks

```
<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" "http://www.w3.org/TR/xhtml11/
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <meta http-equiv="Cache-Control" content="no-cache, no-store, mu
    <meta http-equiv="refresh" content="0; URL=cgi-bin/luci/" />
    <style type="text/css">
      body { background: white; font-family: arial, helvetica,
      a { color: black; }

      @media (prefers-color-scheme: dark) {
        body { background: black; }
        a { color: white; }
      }
    </style>
  </head>
  <body>
    <a href="cgi-bin/luci/">LuCI - Lua Configuration Interface</a>
  </body>
</html>

- /www/index.html 1/20 5%
```

Figure 10: OpenWrt Web Interface HTML in /www/index.html

4.1.3 Configure Firewall Rules

```
C:\Users\admin>ssh root@192.168.56.2
The authenticity of host '192.168.56.2 (192.168.56.2)' can't be established.
ED25519 key fingerprint is SHA256:JU7JpdvC4sg7P+7Cn9xGgQoEdoxf5ZcHrKDepPIu7+U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.2' (ED25519) to the list of known hosts.
root@192.168.56.2's password:
Permission denied, please try again.
root@192.168.56.2's password:
Permission denied, please try again.
root@192.168.56.2's password:
Permission denied, please try again.
root@192.168.56.2's password:
Permission denied, please try again.
root@192.168.56.2's password:
Connection closed by 192.168.56.2 port 22
```

Figure 11: Failed SSH Authentication Attempt on OpenWrt Through TCP Port 22

```

tcp      0      0 0.0.0.0:22          0.0.0.0:*          LISTEN
1691/dropbear
tcp      0      0 :::22              :::*                LISTEN
1691/dropbear
root@OpenWrt:/# uci show dropbear
dropbear.@dropbear[0]=dropbear
dropbear.@dropbear[0].PasswordAuth='on'
dropbear.@dropbear[0].RootPasswordAuth='on'
dropbear.@dropbear[0].Port='22'
root@OpenWrt:/# uci add firewall rule
cfg0f92bd
root@OpenWrt:/# uci set firewall.@rule[-1].name='Allow-SSH-2222'
root@OpenWrt:/# uci set firewall.@rule[-1].src='lan'
root@OpenWrt:/# uci set firewall.@rule[-1].proto='tcp'
root@OpenWrt:/# uci set firewall.@rule[-1].dest_port='2222'
root@OpenWrt:/# uci set firewall.@rule[-1].target='ACCEPT'
root@OpenWrt:/# uci commit firewall
root@OpenWrt:/# /etc/init.d/firewall reload
root@OpenWrt:/# uci show firewall | grep -A6 'Allow-SSH-2222'
firewall.@rule[10].name='Allow-SSH-2222'
firewall.@rule[10].src='lan'
firewall.@rule[10].proto='tcp'
firewall.@rule[10].dest_port='2222'
firewall.@rule[10].target='ACCEPT'
root@OpenWrt:/#

```

Figure 12: OpenWrt Firewall Configuration Allowing SSH Traffic on TCP Port 2222

```

dropbear.@dropbear[0].Port='22'
root@OpenWrt:/# uci add firewall rule
cfg0f92bd
root@OpenWrt:/# uci set firewall.@rule[-1].name='Allow-SSH-2222'
root@OpenWrt:/# uci set firewall.@rule[-1].src='lan'
root@OpenWrt:/# uci set firewall.@rule[-1].proto='tcp'
root@OpenWrt:/# uci set firewall.@rule[-1].dest_port='2222'
root@OpenWrt:/# uci set firewall.@rule[-1].target='ACCEPT'
root@OpenWrt:/# uci commit firewall
root@OpenWrt:/# /etc/init.d/firewall reload
root@OpenWrt:/# uci show firewall | grep -A6 'Allow-SSH-2222'
firewall.@rule[10].name='Allow-SSH-2222'
firewall.@rule[10].src='lan'
firewall.@rule[10].proto='tcp'
firewall.@rule[10].dest_port='2222'
firewall.@rule[10].target='ACCEPT'
root@OpenWrt:/# uci set dropbear.@dropbear[0].port='2222'
root@OpenWrt:/# uci commit dropbear
root@OpenWrt:/# /etc/init.d/dropbear restart
root@OpenWrt:/# netstat -lntp | grep -E ':(22|2222)\b'
tcp      0      0 0.0.0.0:22          0.0.0.0:*          LISTEN
14951/dropbear
tcp      0      0 :::22              :::*                LISTEN
14951/dropbear
root@OpenWrt:/# _

```

```

C:\Users\admin>ssh -p 2222 root@192.168.56.2
ssh: connect to host 192.168.56.2 port 2222: Connection refused

```

Figure 13: SSH Port Modification to TCP 2222 and Connection Refusal During Port Verification

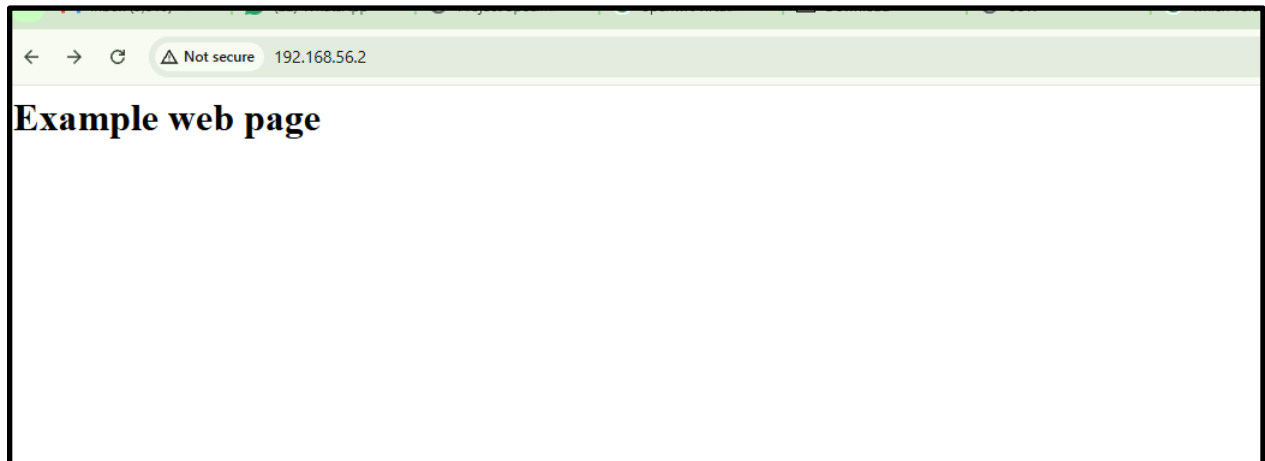


Figure 14: Successful Access to the OpenWrt Web Server Through HTTP

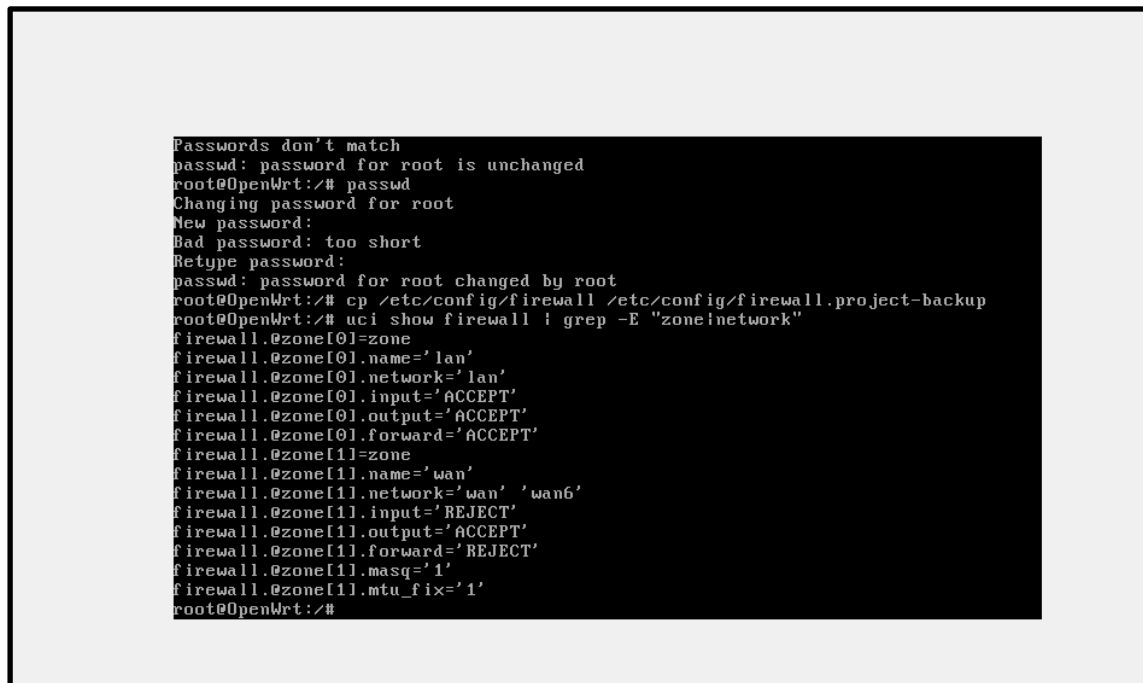


Figure 15: OpenWrt Root Password Hardening and Firewall Zone Configuration

```

uhttpd.main.rfc1918_filter='1'
uhttpd.main.max_requests='3'
uhttpd.main.max_connections='100'
uhttpd.main.cert='/etc/uhttpd.crt'
uhttpd.main.key='/etc/uhttpd.key'
uhttpd.main.cgi_prefix='/cgi-bin'
uhttpd.main.lua_prefix='/cgi-bin/luci=/usr/lib/lua/luci/cgi/uhttpd.lua'
uhttpd.main.script_timeout='60'
uhttpd.main.network_timeout='30'
uhttpd.main.http_keepalive='20'
uhttpd.main.tcp_keepalive='1'
uhttpd.main.ubus_prefix='/ubus'
uhttpd.defaults=cert
uhttpd.defaults.days='730'
uhttpd.defaults.key_type='ec'
uhttpd.defaults.bits='2048'
uhttpd.defaults.ec_curve='P-256'
uhttpd.defaults.country='ZZ'
uhttpd.defaults.state='Somewhere'
uhttpd.defaults.location='Unknown'
uhttpd.defaults.commonname='OpenWrt'
uhttpd.student=uhttpd
uhttpd.student.listen_http='80'
uhttpd.student.home='/srv/www'
root@OpenWrt:~# S

```

Figure 16: OpenWrt HTTP Service Configuration and TCP Port 80 Firewall Rule

```

uhttpd.defaults.ec_curve='P-256'
uhttpd.defaults.country='ZZ'
uhttpd.defaults.state='Somewhere'
uhttpd.defaults.location='Unknown'
uhttpd.defaults.commonname='OpenWrt'
uhttpd.student=uhttpd
uhttpd.student.listen_http='80'
uhttpd.student.home='/srv/www'
root@OpenWrt:~# uci add firewall rule
cfg0e92bd
root@OpenWrt:~# uci set firewall.@rule[-1].name='Block-http-port-80'
root@OpenWrt:~# uci set firewall.@rule[-1].src='lan'
root@OpenWrt:~# uci set firewall.@rule[-1].proto='tcp'
root@OpenWrt:~# uci set firewall.@rule[-1].dest_port='80'
root@OpenWrt:~# uci set firewall.@rule[-1].target='DROP'
root@OpenWrt:~# uci commit firewall
root@OpenWrt:~# /etc/init.d/firewall reload
root@OpenWrt:~# uci show firewall | grep -A6 'Block-HTTP'
root@OpenWrt:~# uci set firewall.@rule[-1].target='ACCEPT'
> ^C
root@OpenWrt:~# uci set firewall.@rule[-1].target='ACCEPT'
root@OpenWrt:~# uci commit firewall
root@OpenWrt:~# /etc/init.d/firewall reload
root@OpenWrt:~# uci show firewall | grep -A6 'Block-HTTP'
root@OpenWrt:~#

```

Figure 17: Successful ICMP Connectivity Test from Windows Host to OpenWrt

```
</html>

C:\Users\admin>ping 192.168.56.2

Pinging 192.168.56.2 with 32 bytes of data:
Reply from 192.168.56.2: bytes=32 time<1ms TTL=64
Reply from 192.168.56.2: bytes=32 time=1ms TTL=64
Reply from 192.168.56.2: bytes=32 time<1ms TTL=64
Reply from 192.168.56.2: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.56.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\admin>
```

Figure 18: OpenWrt Firewall and Network Security Configuration Verification

Firewall Filtering provides some extra security on the network, as it restricts access to key services. The block of port 80 prevents any unauthorised access of web services and then it is opened back to test the service availability in a controlled way. Administrative access through SSH port 22 is initially allowed and it is later switched to port 2222 thus lessening the chances of making SSH automatic attacks (Kuruppathukattil, 2025). Blocking ICMP helps prevent network reconnaissance, re-enabling it when it is needed for connectivity testing. The management interface port 81 is limited and access to it may not be provided to any unauthorised administrative access. Together, these rules mandate controlled connectivity, eases the attack surface, protects administrative services and enhances network security.

4.1.4 Network Diagram and Address Allocation Production

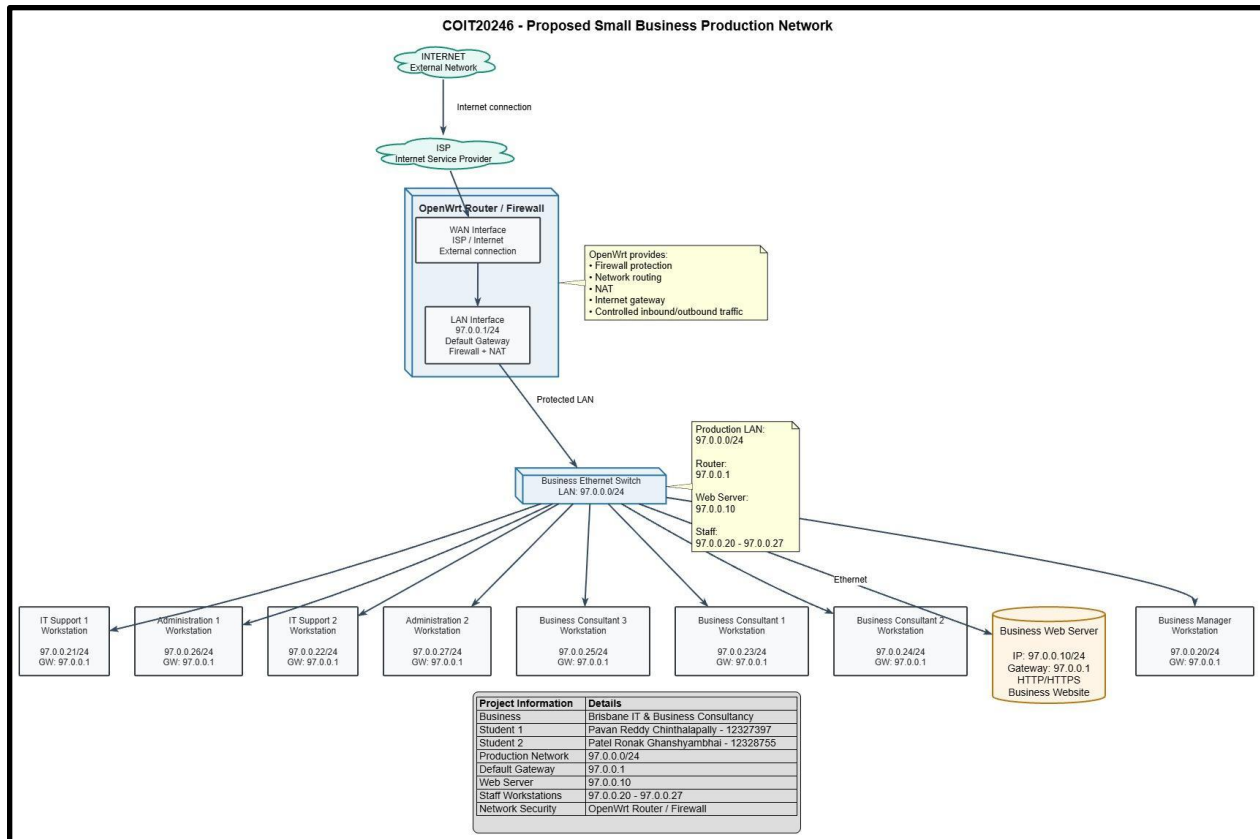


Figure 19: Proposed Small Business Production Network Topology and IP Address Allocation

The suggested production network includes a OpenWrt router/firewall for Internet connection and several routing and NAT services, thus offering traffic protection and routing to the Internet. The protected 97.0.0.0/24 LAN is used to physically connect the business Ethernet switch and the dedicated web server to 8 staff workstations. The web server uses 97.0.0.10, while staff devices use 97.0.0.20–97.0.0.27 (Gentile *et al.* 2024). The default gateway for the router is 97.0.0.1. The structured design facilitates safe connectivity, managed network access, business website functionality and resource management.

4.1.5 IP Addressing Requirements

Device / Role	IP Address	Subnet Mask	Default Gateway	Purpose
OpenWrt Router/Firewall	97.0.0.1	255.255.255.0 (/24)	—	Network gateway, routing and firewall

Business Web Server	97.0.0.10	255.255.255.0 (/24)	97.0.0.1	Business website
Business Manager	97.0.0.20	255.255.255.0 (/24)	97.0.0.1	Staff workstation
IT Support 1	97.0.0.21	255.255.255.0 (/24)	97.0.0.1	Staff workstation
IT Support 2	97.0.0.22	255.255.255.0 (/24)	97.0.0.1	Staff workstation
Business Consultant 1	97.0.0.23	255.255.255.0 (/24)	97.0.0.1	Staff workstation
Business Consultant 2	97.0.0.24	255.255.255.0 (/24)	97.0.0.1	Staff workstation
Business Consultant 3	97.0.0.25	255.255.255.0 (/24)	97.0.0.1	Staff workstation
Administration 1	97.0.0.26	255.255.255.0 (/24)	97.0.0.1	Staff workstation
Administration 2	97.0.0.27	255.255.255.0 (/24)	97.0.0.1	Staff workstation

References

- Kuruppathukattil, V., 2025, August. Optimized WiFi Network Deployment with OpenWRT and FreeRadius for Secure and Scalable Connectivity. In 2025 IEEE 2nd International Conference on Information Technology, Electronics and Intelligent Communication Systems (ICITEICS) (pp. 1-6). IEEE.
- Gentile, A.F., Macrì, D., Greco, E. and Fazio, P., 2024. IoT IP overlay network security performance analysis with open source infrastructure deployment. Journal of Cybersecurity and Privacy, 4(3), pp.629-649.